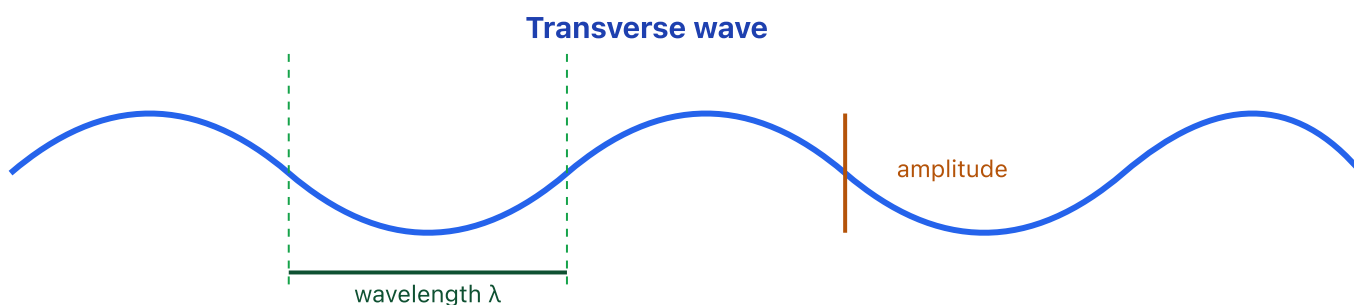


Topic 4 — Waves

Edexcel GCSE (9-1) Physics (1PH0) — Paper 1

4.1 Transverse vs Longitudinal Waves

Waves transfer **energy** (and information) without transferring matter.



Oscillations perpendicular to the direction of energy transfer. Examples: EM waves, water ripples.

Longitudinal wave



4.2 Wave Equation

$$v = f \times \lambda$$

- v — wave speed (m/s)
- f — frequency (Hz = waves per second)
- λ — wavelength (m)

$$T = 1 / f \quad (\text{period} = 1 / \text{frequency})$$

Worked Example: A sound wave has frequency 500 Hz in air ($v = 340$ m/s).

$$\lambda = v / f = 340 / 500 = \mathbf{0.68 \text{ m}}.$$

4.3 Wave Behaviour at Boundaries

- **Reflection** — bounces off (angle of incidence = angle of reflection).
- **Refraction** — changes speed and direction on entering a new medium; frequency unchanged.
- **Absorption** — wave energy transferred into material.
- **Transmission** — passes through without significant loss.

4.4 Sonar and Ultrasound

distance = speed $\times$ time (then halve, because the wave goes there and back)

Worked Example: A ship's sonar echo returns in 0.40 s ($v_{\text{sound in water}} = 1\,500$ m/s).

$$\text{Total path} = 1\,500 \times 0.40 = 600 \text{ m} \rightarrow \text{depth} = \mathbf{300 \text{ m}}.$$

Ultrasound (>20 kHz) is used in medicine because it is non-ionising and gives real-time images of soft tissue — e.g. fetal scans.

4.5 Seismic Waves

- **P-waves:** longitudinal; travel through solids *and* liquids.
- **S-waves:** transverse; cannot travel through liquids.

S-wave shadow on the far side of the Earth is strong evidence for a **liquid outer core**.

Key Points Summary

- Transverse oscillations $\perp$ direction of travel; longitudinal oscillations $\parallel$ direction of travel.
- $v = f\lambda$ and $T = 1/f$ — learn and rearrange.
- Reflection, refraction, absorption, transmission are the four behaviours at boundaries.
- Ultrasound in medicine is safe, non-ionising; sonar uses reflection to measure distance.
- P-waves travel through everything; S-waves only solids — evidence for Earth's liquid core.